

Statistics, Counting & Probability



The fundamental counting principle

- 1 A restaurant menu has 4 appetizers, 6 main courses, and 3 desserts. How many different 3-course meals are possible?
 - 2 A license plate consists of 3 letters followed by 4 digits. How many different license plates are possible, assuming repetition is allowed?
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Permutations

- 3 In how many ways can 5 different books be arranged on a shelf?
 - 4 In how many ways can a president, vice president, and secretary be chosen from a club of 10 members?
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Combinations

- 5 In how many ways can a hand of 5 cards be dealt from a standard 52-card deck?
 - 6 A committee of 4 is chosen from 6 men and 5 women. In how many ways can the committee consist of exactly 2 men and 2 women?
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Probability with combinations

- 7 A bag contains 5 red and 7 blue marbles. If 3 marbles are drawn at random without replacement, find the probability all 3 are red.
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The binomial probability formula

- 8 A fair coin is flipped 5 times. Find the probability of getting exactly 3 heads, using the binomial probability formula $P(k) = \binom{n}{k} p^k (1 - p)^{n-k}$.
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Expected value and variance

- 9 A game gives a payout of 10 dollars with probability 0.3, 5 dollars with probability 0.4, and 0 dollars with probability 0.3. Find the expected payout.

Normal distribution and z-scores

- 10 A data set is normally distributed with mean 70 and standard deviation 8. Find the z-score for a data value of 86.
- 11 Using the empirical rule (68-95-99.7 rule), estimate the percentage of data within 2 standard deviations of the mean.

A multi-part probability and counting problem

- 12 This question has two parts. A class has 8 boys and 7 girls. A group of 4 students is selected at random to represent the class. (a) Find the total number of ways to select the group of 4 from the 15 students. (b) Find the probability that the group consists of exactly 2 boys and 2 girls.

Permutations with repeated elements

- 13 Find the number of distinct arrangements of the letters in the word "BANANA".

Conditional probability with counting

- 14 A committee of 3 is selected from 5 men and 4 women. Given that at least one woman is selected, find the probability that exactly 2 women are selected.

Percentiles and standardized scores

- 15 A student scores at the 85th percentile on a standardized test. Explain what this means in terms of the rest of the test-taking population.

Mutually exclusive vs. overlapping events

- 16 A die is rolled once. Find $P(\text{rolling an even number or a 5})$, explaining why the addition rule simplifies since these events are mutually exclusive.
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A multi-part counting and probability problem with restrictions

- 17** This question has three parts. A password consists of 4 characters chosen from the 10 digits 0-9, and repetition is not allowed. (a) Find the total number of possible passwords. (b) Find the number of passwords where the first character is not 0 (a common restriction to avoid leading zeros). (c) Find the probability that a randomly generated valid password (from part b) starts with an even digit.